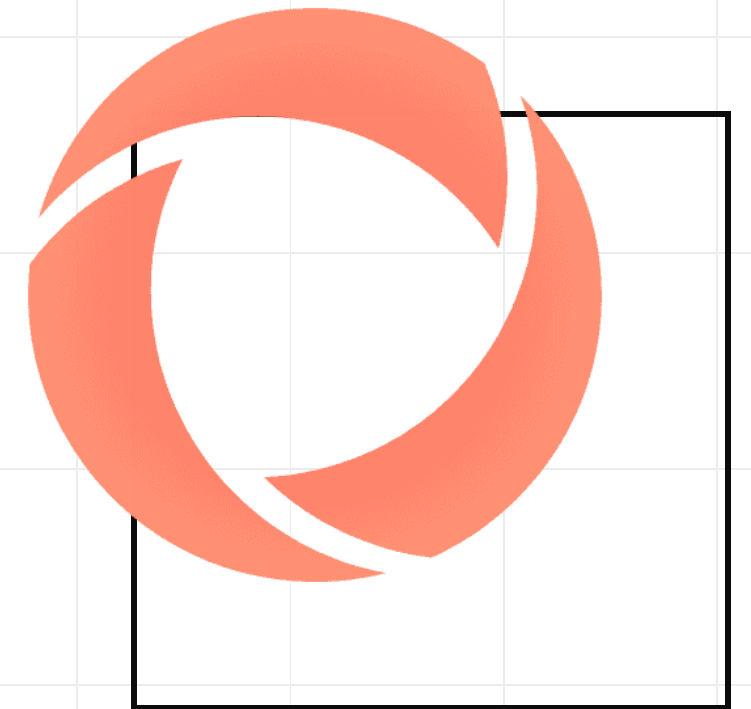


A GUIDED TOUR OF THE FRAMEWORK

Qensei

The AI-assistant-driven QA framework. From a manually-tested ticket to a self-cleaning, REST-first regression pack — with an advisory review panel that raises the floor.



WHAT IT IS

A backend-aware QA engine, driven by an AI **sensei**

02

Qensei = Q (QA) + **sensei**. An AI assistant drives it like a QA sensei — **reading** the backend to design cases, **verify** tickets and **diagnose** failures. It analyses and verifies work; it never codes product features. Three capabilities that usually live in separate tools, unified under one product-neutral engine.

Test-case design

Manual-QA case design **with domain knowledge** of the product under test.

REST regression

Automated regression suites run against a live backend — the deterministic gate.

Failure diagnostics

Classify a failure by reading the backend contract, not by guessing.

THE PROBLEM IT SOLVES

Three problems, **one** design response

03

01

Fragmentation

Manual test design, automated regression, and diagnostics live in three separate tools.

02

Product lock-in

Most QA tooling is coupled to one product. Adding a product means rewriting the tool.

03

The AI-test trap

AI automation passes every mock-level check while silently failing the real boundary.

So – **who owns “green”** when an AI writes the tests? Qensei answers it structurally: the gate owns green, the human owns convergence, the AI owns implementation.

THE MENTAL MODEL

Three capabilities over **one** backend connection

04

ONE BACKEND CONNECTION · the SUTConnector

reads the SOURCE — or the ticket + docs when a SUT is sourceless — and drives the RUNTIME

**DESIGN**

Reads the backend surface — endpoints + business rules — and reports coverage gaps and candidate cases.

`make design`

**REGRESS**

Runs the pack suite against a live backend. This is the deterministic gate — the source of truth for green.

`make test`

**DIAGNOSE**

Classifies a failure as REAL_BUG vs TEST_BUG by reading the backend contract.

`make diagnose`

Qensei **reads and exercises** the backend for QA — to design cases, verify tickets and diagnose failures. It never modifies the product or ships features.

THE DETERMINISTIC CORE

The gate owns “green” — **no AI in it**

05

The assistant designs, validates, and diagnoses **around** the gate — it never becomes the gate.

```
engine/ · make test · make demo — plain Python 3 stdlib, the single source of truth for green
```

Deterministic core

run.py runs every pack against the SUT and exits non-zero if any case fails. Zero-dependency. No AI in the loop.

The AI assistant

Slash commands and the review panel work **around** the gate — they accelerate the work, but the human owns convergence.

THE OWNERSHIP MODEL

Who owns what

06

HUMANS OWN INTENT Intent and design convergence. The spec — the what and the why — is human-approved.	THE AI OWNS THE QA BUILD Designs, validates and diagnoses the tests around the gate — not product features. The plan — the how — is agent-iterated.	THE GATE OWNS “GREEN” Deterministic, zero-dependency, no AI. It is the only thing that decides pass or fail.
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Humans own intent and design convergence; the AI owns **implementation**.

THE END-TO-END FLOW

From a manually-tested ticket to a **pack**

07

/validate

Check a ticket's acceptance criteria against a live SUT. Output: a run report with evidence.



/automate

Turn that validated result into an intent spec + a permanent regression pack.



/report-bug

When triage finds a real defect, file a structured Bug — human-gated.

```
mock-file:SHOP-456 → specs/SHOP-456-bulk-discount.md → packs/SHOP-456-discount/ — the full, runnable entry flow.
```

Validate a ticket against a live SUT

The manual-validation leg. Its output — **a run report with evidence** — is the source of truth a `/automate` run turns into an automated regression.

- 1

Setup

Read policies + the active SUT's knowledge; fetch & normalize the ticket; triage depth.
- 2

Environment

Select an env from the manifest; collect version info; mask secrets.
- 3

Validation

Each AC becomes a check — REST or UI. UI runs **headed** so a QA person watches.

- 4

Write results

Record outcomes back on the ticket — approval-gated.
- 5

Transition

Move the ticket forward only if every check passes.
- 6

Report

Save a run report to `runs/<TICKET>_<TS>.md` and append a learning.

Turn the validated result into a permanent pack

REST-first, UI fallback — an intent spec plus a regression pack, validated against the gate. **The surface follows the verification.** The ticket (incl. its **comments**) can even come pre-validated **outside** Qensei — a prior `/validate` run isn't required.



THE SPINE OF THE FRAMEWORK

Never weaken a test to **turn it green**

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The validation objective **never** changes during Phase 4. Every fix preserves the acceptance criteria verbatim.

Forbidden without re-approval

Relax = 100 to ≥ 90 · drop a persona marker · add an ungated xfail · raise a timeout to silence a real regression.

Loop budget

After 3 cycles on the same root cause — **STOP and escalate**. The panel mirrors it: twice the same finding → stop.

THE SUT PLUGIN MODEL

The product under test is a **plugin**

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A “**site**” under `sut/<name>/` owns both its backend access and its tests. The engine depends only on this contract, **never on a specific product** — you wire in **your own** product as a plugin. The three SUTs in the repo are **worked examples**.

manifest.json declares

`source (path · repo · ref)` — or omit it for a **sourceless SUT** · `tests · runtime (in_process / remote) · env · creds · knowledge`.

Real source, kept fresh

Point `source.repo` at the backend and make `sync-source` keeps a **local clone** of the real code current — for design, diagnosis and citations. A **sourceless** SUT has no clone to keep fresh — its **ticket + docs** are the contract of record.

Two more sites — a fuller mock and a **sourceless** plugin (no readable source) — drop in with **no change to engine/ or policies/**. That is what proves the seam is genuinely generic.

PER-SUT KNOWLEDGE

skills vs learnings — loaded on demand

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knowledge.skills

Authored, standing domain knowledge — what the product does, where contracts live, the gotchas. e.g. SHOP.md, BOOKER.md.

knowledge.learnings

Append-only, dated, AI-accumulated patterns — one captured per Phase-4 cycle as the framework works.

Loaded on demand by `--sut` — only the **active** product's knowledge enters context.

THREE EXAMPLE SUTS SHIP WITH THE FRAMEWORK — DEMOS THAT PROVE THE SEAM, NOT REAL PRODUCTS

mock-shop — example · in_process · REST-only · 10% off at ≥3 items

restful-booker — example · plugin.py + ui-packs + a live remote env · HTTP 409 no-double-booking

widget-api — example · sourceless · remote runtime, no source · ticket + skills are the contract

THE REVIEW PANEL

Seven advisory lenses

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judge

Presides — BLOCK / FIX / FLAG / ESCALATE; writes the human digest.

r-diagnosis

TEST_BUG vs REAL_BUG — before any fix, with cited evidence.

r-evidence

The skeptic. **The green dot is NOT evidence** — pull the raw state.

r-mechanism

SUT-behaviour claims — cite `source:line` (or a ticket/doc anchor) or escalate.

r-fidelity

Catches edits that **weaken** an AC to turn a red test green.

r-coverage

Flags an AC the pack never exercises, or a `covers/contract_claim` naming no real rule.

r-uplift

Migration-only — did a ported legacy test adopt these patterns?

The panel **raises the floor** — it never lowers the bar, and it never blocks a merge. Read-only is tool-enforced (**gate: false**).

HOW THE REPO IS LAID OUT

One core, many plugins

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```
commands/  # slash commands – the human-in-the-loop legs
agents/    # the advisory review panel (read-only)
policies/  # 8 product-neutral QA governance policies
engine/    # the deterministic core – owns "green"
sut/       # the SITES under test – plug in YOUR product
├─ mock-shop/      # example plugin (demo)
├─ restful-booker/ # example plugin (demo)
└─ widget-api/     # example · sourceless (no source/)
ticket/      # tracker-agnostic ticket contract
docs/        # overview + architecture + gates
tests/       # site-integration bridge (pytest)
```



Two plugin seams

Backend access (sut/contract.md) and ticket access (ticket/contract.md).



Swap either seam

...and **nothing** in engine/ or policies/ changes.

QENSEI, IN ONE LINE

One engine · a plugin per product · a panel that **raises the floor**

One backend-aware QA engine with **three capabilities** over a single backend connection, the product under test wired in as a **plugin**, an **advisory panel** that raises the floor — and a **deterministic gate** that owns “green.”

START HERE

\$ make demo

\$ make demo-booker

\$ make design

\$ make test

